

(MODEL)

TED (15) 3134

Regno

(REVISION 2015)

Signature

THIRD SEMESTER DIPLOMA EXAMINATION IN INFORMATION TECHNOLOGY

OBJECT ORIENTED PROGRAMMING THROUGH C++

[Time : 3 hours

(Maximum Marks : 100)

PART – A

(Maximum marks : 10)

Marks

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Write the syntax of function prototype.
2. State the use of setw manipulator.
3. Define Encapsulation.
4. State the advantage of Inheritance
5. Specify the use of virtual functions

PART – B

(Maximum marks : 30)

II. Answer *any five* of the following. Each question carries 6 marks

1. Write a program to search an element in a one dimensional array.
2. Explain about heterogeneous aggregate.
3. Explain Function overloading
4. Illustrate the relationship between a class and object with an example
5. Explain about friend function.
6. Explain about visibility controls
7. Explain with example how function templates are implemented.

PART – C

(Maximum marks : 60)

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT - I

III. (a) Explain about different operators in C++ (9 marks)

(b) Compare while and do – while loops (6 marks)

OR

- IV. (a) Explain the representation of strings in C++ and various string operations. (9 marks)
(b) Explain the primitive data types in C++ (6 marks)

UNIT - II

- V. (a) Explain different parameter passing methods (9 marks)
(b) List the features of object oriented programming (6 marks)

OR

- VI. (a) Develop class Employee with member variable Employee No (type int), employee name (type String) and member functions to read and display the features of an employee. Write constructors initialize the member variable. (9marks)
(b) Illustrate the use of static data member in class with an example. (6 marks)

UNIT - III

- VII. (a) Explain operator overloading (9 marks)
(b) Illustrate how constructors of base class can be invoked from a derived class (6 marks)

OR

- VIII. Create a class publication that stores the title (a string) and price (type float) of a publication. From this class derive two classes: book. which adds a page count (type int), and tape, which adds a playing time in minutes (type float). Each of these three classes should have a getdata() function to get its data from the user at the keyboard, and a putdata() function to display its data. Write a main() program to create objects of book and tape classes and display the output. (9 marks)
(b) Explain limitations operator overloading. (6 marks)

UNIT - IV

- IX. (a) Explain exception handling in C++ (9 marks)
(b) Write about pure virtual function (6 marks)

OR

- X. (a) Write a program that reads a group of numbers from the user and places them in an array of type int. The program should find the sum of numbers in the array using a function sum() which takes the array as argument. Uses pass by pointers. (9 marks)
(b) Write a short note on multiple inheritance (6 marks)